Finite-radius rank-two local optimality for a qubit quantum-embeddability ansatz

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Abstract

Classical transition matrices obtained by preparing and measuring a qubit in one fixed basis form a constrained subset when the intervening evolution is a time-homogeneous quantum dynamical semigroup. A recent numerical study proposed that a four-parameter Hamiltonian–one-jump family generates the boundary of the qubit set. We do not resolve that global question. Instead, we prove a computer-assisted finite-radius result at one stationary point of the section $a = 1/2$. In an explicit eleven-coordinate, gauge-fixed canonical chart for Kossakowski rank at most two, an interval-Newton enclosure fixes the one-jump point and outward-rounded matrix-exponential derivatives bound the Hessian on the complete cube $|y_i| \le 10^{-4}$. The smallest directed Gershgorin lower endpoint is larger than 0.0333886, which implies

$$b(y) - b_\star \ge 0.0165\,\|y\|_2^2 \quad \text{whenever} \quad a(y) = 1/2.$$

Here $b_\star = 0.02261621659234744314\ldots$. A 320-bit replay and a separately constructed complex matrix-unit calculation provide independent diagnostics. Thus a genuinely transverse second dissipative amplitude cannot improve the tradeoff inside this normalized neighborhood. The statement excludes rank-three generators, distant rank-two charts, global optimization, unbounded parameter tails, and any identification of the true boundary.

1 Introduction

The embedding problem asks whether a stochastic transition matrix can arise from continuous-time memoryless dynamics. Allowing a finite-dimensional quantum system between classical preparation and measurement enlarges the set of possible stochastic matrices and leads to the notion of quantum embeddability [1, 2]. For a two-level system, Shahbeigi *et al.* obtained analytical inclusions and exclusions and compared a general numerical search with a four-parameter family containing one rank-one jump operator [2]. Their agreement at entrywise tolerance 10^{-4} motivated the open question of whether that family generates the boundary of the closed qubit set.

Numerical agreement does not by itself control local optimizer failure, unbounded rate and time limits, or additional dissipative directions. The full boundary question therefore naturally separates into local and global parts. This paper addresses one local part: can a second, genuinely independent jump lower the survival probability b at fixed $a = 1/2$ near a specific stationary point of the one-jump ansatz?

Our contribution is an explicit finite-amplitude answer within Kossakowski rank at most two. We construct a canonical chart that removes jump mixing, jump phase, and the fixed-basis system rephasing at the rank-one point. The second jump enters this chart quadratically, so pointwise first-order tests cannot settle the question. We instead enclose the full Hessian of a penalized support functional throughout a normalized coordinate cube using Arb ball arithmetic [3]. Strong convexity on the cube then gives the stated fixed-section inequality by an integral remainder

argument. The assumptions and the exact scope are collected in Theorem 1 within the first two sections.

The result is deliberately not presented as a boundary theorem. It neither proves that the base point lies on the true quantum-embeddable boundary nor that it is the global minimum even within the original four-parameter family. Its value is narrower: any rank-two counterexample at $a = 1/2$ must leave the displayed local chart cube. A bounded literature search through 12 September 2026 located no prior finite-radius statement of this form, but that non-detection is not a priority guarantee.

2 Model, observable, and the one-jump point

2.1 Time-homogeneous qubit dynamics

Let ρ be a positive semidefinite 2×2 matrix of unit trace. A time-independent Markovian evolution has a Gorini–Kossakowski–Sudarshan– Lindblad generator [4, 5]

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_k \mathcal{D}[A_k](\rho), \qquad \mathcal{D}[A](\rho) = A\rho A^\dagger - \frac{1}{2}\{A^\dagger A, \rho\}. \tag{1}$$

Preparation and measurement use the same fixed basis $\{|0\rangle, |1\rangle\}$. Absorbing the finite evolution time into the generator, define

$$T_{ij} = \langle i|e^{\mathcal{L}}(|j\rangle\langle j|)|i\rangle, \qquad T(a, b) = \begin{pmatrix} a & 1-b \\ 1-a & b \end{pmatrix}. \tag{2}$$

Thus $a = T_{00}$ and $b = T_{11}$ are dimensionless survival probabilities. This convention describes each finite generator. The closed set Q_2 in Ref. [2] additionally permits limits of sequences whose times or rates are not uniformly bounded; no such limiting argument is used below.

Write $\rho = (I + \boldsymbol{r}\cdot\boldsymbol{\sigma})/2$. For traceless jumps $A_k = \boldsymbol{z}_k \cdot \boldsymbol{\sigma}$, with $\boldsymbol{z}_k = \boldsymbol{u}_k + i\boldsymbol{v}_k$, the affine Bloch equation is

$$\begin{aligned} \dot{\boldsymbol{r}} &= M\boldsymbol{r} + \boldsymbol{c}, \\ M &= 2[\boldsymbol{h}]_\times + 2\sum_k \left(\boldsymbol{u}_k\boldsymbol{u}_k^\mathsf{T} + \boldsymbol{v}_k\boldsymbol{v}_k^\mathsf{T} - (\|\boldsymbol{u}_k\|^2 + \|\boldsymbol{v}_k\|^2)I_3\right), \quad \boldsymbol{c} = 4\sum_k \boldsymbol{u}_k \times \boldsymbol{v}_k, \end{aligned} \tag{3}$$

where $H = \boldsymbol{h}\cdot\boldsymbol{\sigma}$. If $e^{\widetilde{M}} = \left(\begin{smallmatrix} R & \boldsymbol{d} \\ 0 & 1 \end{smallmatrix}\right)$ for $\widetilde{M} = \left(\begin{smallmatrix} M & \boldsymbol{c} \\ 0 & 0 \end{smallmatrix}\right)$, then

$$a = \frac{1 + R_{zz} + d_z}{2}, \qquad b = \frac{1 + R_{zz} - d_z}{2}. \tag{4}$$

The calculation evaluates this full exponential; there is no small-time or perturbative truncation.

2.2 Restricted stationary point

The family proposed in Appendix A of Ref. [2] is

$$\begin{aligned} H_4 = \sigma_x, \qquad A = \sqrt{\gamma}\,|\psi_o(\alpha)\rangle\langle\psi_i(\beta)|, \\ |\psi_o(\alpha)\rangle = (\cos\alpha, i\sin\alpha)^\mathsf{T}, \qquad |\psi_i(\beta)\rangle = (\cos\beta, i\sin\beta)^\mathsf{T}, \end{aligned} \tag{5}$$

with $\gamma \geq 0$ and finite time $t > 0$. At fixed $a = 1/2$, let $p = (t, \gamma, \alpha, \beta)$ and solve the five Karush–Kuhn–Tucker equations

$$a(p) - \tfrac{1}{2} = 0, \qquad \nabla_p\big[b(p) - \lambda a(p)\big] = 0. \tag{6}$$

Interval Newton proves that the box of componentwise radius 10^{-3} about

$$\begin{aligned} p_0 &= (1.6856350198672431,\ 0.8285640995933057,\\ &\qquad 0.3147912138228343,\ -0.4934749196785660),\\ \lambda_0 &= 0.03174580990704283 \end{aligned} \tag{7}$$

contains a unique solution $(p_\star, \lambda_\star)$ of Eq. (6). The much tighter propagated enclosure gives

$$\begin{aligned} b_\star \in [&0.022616216592347443142445095482107697263874053954793357009735228 99,\\ &0.022616216592347443142445095482107697263874053954793357009739240 48]. \end{aligned} \tag{8}$$

To express this point in a canonical traceless-jump gauge, set $q = \sqrt{t\gamma}$, $d = \alpha - \beta$, and $s = \alpha + \beta$. Removing the identity part of the jump together with its Hamiltonian compensation gives

$$h_x = t - \frac{t\gamma}{8}\sin(2d), \qquad \boldsymbol{z} = \left(\frac{iq\sin d}{2}, \frac{q\sin s}{2}, \frac{q\cos s}{2}\right). \tag{9}$$

At the solution, the rounded native coordinates are

$$\begin{aligned} (h_x, h_y, h_z, \Re z_y) &= (1.511235499, 0, 0, -0.105023456),\\ (\Re z_z, \Im z_x, \Im z_y) &= (0.581493112, 0.427275058, 0). \end{aligned} \tag{10}$$

All certification uses the interval enclosures, not the rounded numbers in Eqs. (7) and (10).

3 Canonical rank-two chart and main result

Consider

$$\mathcal{L}_{\boldsymbol{h},\boldsymbol{z},\boldsymbol{w}}(\rho) = -i[\boldsymbol{h}\cdot\boldsymbol{\sigma}, \rho] + \mathcal{D}[\boldsymbol{z}\cdot\boldsymbol{\sigma}](\rho) + \mathcal{D}[\boldsymbol{w}\cdot\boldsymbol{\sigma}](\rho). \tag{11}$$

We use the native real coordinates

$$x = (h_x, h_y, h_z, \Re z_y, \Re z_z, \Im z_x, \Im z_y, \Re\eta_1, \Re\eta_2, \Im\eta_1, \Im\eta_2) \in \mathbb{R}^{11} \tag{12}$$

and define

$$\begin{aligned} \boldsymbol{z} &= (i\Im z_x,\ \Re z_y + i\Im z_y,\ \Re z_z),\\ \boldsymbol{w} &= \left(\eta_1, \eta_2, -\frac{\overline{z}_x\eta_1 + \overline{z}_y\eta_2}{\overline{z}_z}\right). \end{aligned} \tag{13}$$

Consequently $\boldsymbol{z}^\dagger\boldsymbol{w} = 0$ identically and

$$C = \boldsymbol{z}\boldsymbol{z}^\dagger + \boldsymbol{w}\boldsymbol{w}^\dagger \succeq 0, \qquad \operatorname{rank} C \le 2. \tag{14}$$

The certified cube keeps $\Re z_z > 0.5813346$, so the rational coordinate in Eq. (13) is regular.

This chart is locally complete for canonical Kossakowski matrices of rank at most two near the simple rank-one base point. Indeed, spectral amplitudes of a nearby positive semidefinite C give orthogonal $\boldsymbol{z}$ and $\boldsymbol{w}$. The leading-jump phase and the only system-basis diagonal rephasing relevant to the observable fix $\Im z_z = 0$ and $\Re z_x = 0$ on a transverse slice. The nonzero z_z then yields Eq. (13). Jump mixing has been removed by orthogonality; the remaining phase of $\boldsymbol{w}$ does not change $\boldsymbol{w}\boldsymbol{w}^\dagger$.

Let $x_\star$ be Eq. (9) at $p_\star$, padded by $\eta_1 = \eta_2 = 0$. Let $B \in \mathbb{R}^{11\times 11}$ be the fixed invertible whitening matrix tabulated in Appendix A; its full binary64 decimal representation is supplied as `whitening-matrix.csv`. Set

$$x(y) = x_\star + By, \qquad Y_r = \{y \in \mathbb{R}^{11} : |y_i| \le r \text{ for all } i\}. \tag{15}$$

Theorem 1 (Finite-radius rank-two local optimality)**.** *For every $y \in Y_{10^{-4}}$, Eq. (15) defines a finite time-independent qubit GKSL generator whose Kossakowski rank is at most two. If its fixed-basis transition matrix satisfies $a(y) = 1/2$, then*

$$b(y) - b_\star \geq 0.0165\, \|y\|_2^2. \tag{16}$$

In particular, $y = 0$ is the unique minimizer of b on the exact fixed-a section inside this chart cube.

The normalization in Eq. (15) is part of the theorem. The radius 10^{-4} is not an invariant distance in generator space, and the theorem does not compare with rank-three or distant rank-two generators.

4 Computer-assisted proof

4.1 Stationary augmented support functional

Fix the interval-enclosed multiplier $\lambda_\star$ and define

$$F(y) = b(y) - \lambda_\star a(y), \qquad G(y) = F(y) + \big(a(y) - \tfrac{1}{2}\big)^2. \tag{17}$$

The seven derivatives of F on the one-jump gauge slice vanish at the exact root of Eq. (6). The four derivatives in the η coordinates also vanish: $\mathcal{D}[\boldsymbol{w} \cdot \boldsymbol{\sigma}]$ is quadratic in $\boldsymbol{w}$ and $\boldsymbol{w} = 0$ at the base point. Since $a(0) = 1/2$, the penalty has zero gradient there. Hence

$$\nabla_y G(0) = 0 \tag{18}$$

exactly; this conclusion does not replace a zero equation by a floating-point residual.

4.2 Outward-rounded Hessian enclosure

The implementation propagates value, first derivative, and second derivative jets through the rational chart and the affine Bloch generator Eq. (3). Derivatives of the finite-time channel are evaluated through full block-matrix exponentials. At first order this is the Fréchet map

$$De^M[E] = \int_0^1 e^{(1-s)M} E e^{sM}\, \mathrm{d}s, \tag{19}$$

used in matrix-exponential algorithms [6]; second derivatives use the corresponding repeated block construction in the certificate. There is no polynomial truncation of e^M. Every scalar and matrix operation is performed with outward-rounded Arb balls at 256-bit precision.

The fixed matrix B whitens the midpoint of $\nabla_x^2 G(x_\star)$. Direct interval Gershgorin bounds are then applied to the symmetric interval matrix $H(y) = \nabla_y^2 G(y)$ over the complete cube $Y_{10^{-4}}$. The eleven directed row lower endpoints are

$$\begin{aligned}(&0.0333886328,\ 0.8484794697,\ 0.8331210866,\ 0.9108255037,\\ &0.9108255035,\ 0.8116418982,\ 0.9121361431,\ 0.8416532932,\\ &0.8577484555,\ 0.9548488791,\ 0.9548488791).\end{aligned} \tag{20}$$

Thus, throughout the cube,

$$H(y) \succeq 0.033\, I_{11}. \tag{21}$$

The radius trials are summarized in Fig. 1. A negative Gershgorin endpoint at a larger radius only means that this direct enclosure test is inconclusive there.

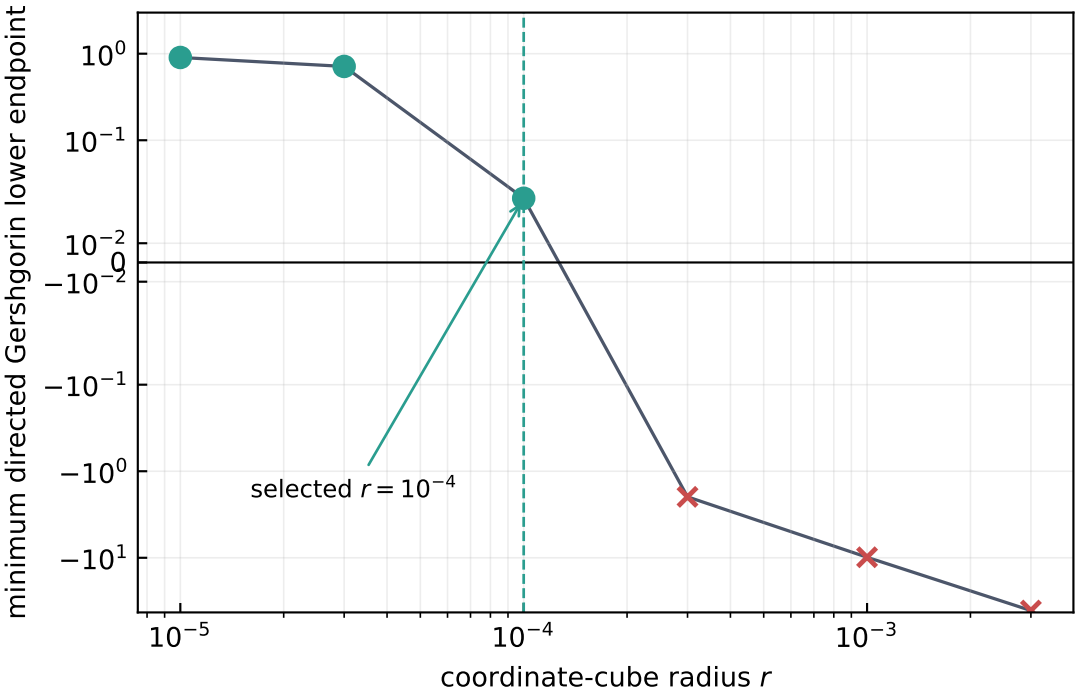


Figure 1: Minimum directed Gershgorin lower endpoint for the interval Hessian at each tested coordinate-cube radius. Filled circles denote a positive uniform certificate and crosses denote an inconclusive direct enclosure. The largest tested radius passing this criterion is 10^{-4}. The plot is regenerated directly from the frozen interval JSON record.

For any $y \in Y_{10^{-4}}$, the segment sy remains inside the cube. Using Eqs. (18) and (21),

$$\begin{aligned} G(y) - G(0) &= \int_0^1 (1-s) y^{\mathsf{T}} H(sy) y \, \mathrm{d}s \\ &\geq 0.0165 \, \|y\|_2^2. \end{aligned} \tag{22}$$

On the exact section $a(y) = a(0) = 1/2$, both penalty terms vanish and the support terms cancel, so $G(y) - G(0) = b(y) - b_\star$. This proves Theorem 1.

4.3 Error controls and independent diagnostics

Three additional checks probe distinct failure modes.

First, repeating the complete interval calculation at 320 bits selects the same radius and preserves every positive row bound. Precision agreement is not used as a substitute for outward rounding; both runs are recorded as ball calculations.

Second, a separate Taylor estimate bounds the Frobenius norm of the full third-derivative tensor of G by 3.291×10^{14} on the containing cube. Combined with a point-Hessian lower bound greater than 0.9999999999999958 in whitened coordinates, it certifies the much smaller radius 9.11×10^{-15}. This deliberately coarse estimate is an independent analytic remainder check; the 10^{-4} theorem instead uses the stronger uniform-Hessian enclosure.

Third, an implementation constructed directly from the action of Eq. (1) on column-major matrix units, rather than from the Bloch generator, finite-differences the whitened Hessian and

corrects random directions back to $a = 1/2$. It finds

$$\begin{aligned} \max_{ij} |H^{\mathrm{MU}}_{ij} - \delta_{ij}| &= 6.60 \times 10^{-6}, \\ \lambda_{\min}(H^{\mathrm{MU}}) &= 0.9999998712, \\ \max |\boldsymbol{z}^{\dagger}\boldsymbol{w}| &= 8.56 \times 10^{-22}, \\ \max |a - 1/2| &= 5.56 \times 10^{-16}. \end{aligned} \tag{23}$$

All sixteen corrected fixed-a trials lie above $b_\star$, with minimum $b - b_\star = 7.996 \times 10^{-10}$. These are diagnostic floating-point tests, not ingredients of the interval proof.

Before the new calculation, two separately derived representations also reproduced the source model. At $(\alpha, \beta, \gamma, t) = (0.2, 1.0, 0.7, 0.9)$, the matrix-unit and affine-Bloch implementations agree in (a, b) within 4.45×10^{-16}. For $\gamma = 0$, both recover $a = b = \cos^2 t$, including the swap at $t = \pi/2$, to 2.23×10^{-16}. This guards the convention map in Eq. (4); it does not establish global optimization.

5 Physical meaning and limitations

At a rank-one Kossakowski point, a second jump amplitude changes the generator only at quadratic order. The chart in Eq. (13) removes the part that merely remixes the existing jump and retains a genuinely transverse dissipative amplitude. Theorem 1 therefore says that, throughout the stated normalized neighborhood, opening such a second channel cannot decrease the fixed-basis survival probability b while preserving $a = 1/2$. At the edge $\|y\|_2 = 10^{-4}$, the certified lower increase is 1.65×10^{-10}; the small numerical scale reflects the local radius, not a perturbative approximation.

The following exclusions are essential.

- The proof treats only the interval-enclosed KKT point at $a = 1/2$ and the explicit chart $Y_{10^{-4}}$.
- Kossakowski rank three is absent. A rank-three direction is not a limit covered by the rank-two chart argument.
- No distant rank-two chart, global minimum of the four-parameter family, or unbounded time/rate tail is controlled.
- The result does not prove that $(1/2, b_\star)$ lies on the boundary of Q_2, nor does it identify any component of that boundary.
- The GKSL equation is an assumed effective model for one qubit. No ancilla, switching, time-dependent control, postselection, or microscopic bath statement is included.

Consequently the original boundary-generation question remains unresolved. The present theorem narrows a local counterexample search; it is not a proof of the one-jump boundary conjecture.

6 Reproducibility and data availability

The accompanying source bundle contains the 256-bit interval program, its KKT and full-slice dependencies, the 320-bit replay, the independent matrix-unit verifier, machine-readable JSON outputs, the exact whitening matrix, and regeneration instructions in `REPRODUCIBILITY.md`. The primary certificate uses python-flint 0.9.0/Arb at 256 bits. The distributed figure is generated by `make_figure.py` from the primary JSON record. No downloaded third-party paper or upstream code is redistributed in the submission bundle; the source article and software are identified by their published references. Public archival location and licensing of the project-generated code remain decisions for the author before submission.

Author contributions and AI-use disclosure

Xiaojian Zeng supplied the research direction and is the prospective author of this draft. Generative-AI systems operating through OpenAI Codex assisted with literature search, formulation of the local claim, derivations, production and execution of calculation code, independent computational checks, figure production, and manuscript drafting. The prospective author has not yet confirmed the scientific argument, code, citations, affiliation, or submission decision and must review the complete work and accept full responsibility before any submission.

Competing interests

No competing-interest statement has yet been confirmed by the prospective author.

A Whitening matrix

The rows of B follow the native-coordinate order in Eq. (12); columns multiply $(y_1, \ldots, y_{11})$. For readability the display rounds entries smaller than 10^{-12} to zero and the others to six significant digits. The theorem and interval calculation use the full binary64 values reproduced in `whitening-matrix.csv` and in the primary JSON record.

$$B \simeq \begin{pmatrix}
0 & 0 & -0.636222 & 0 & 0 & 0.499244 & 0 & 0.541959 & -0.0692943 & 0 & 0 \\
-14.4300 & 0.119842 & 0 & 0 & 0 & 0 & -0.202815 & 0 & 0 & 0 & 0 \\
-1.00410 & 1.78863 & 0 & 0 & 0 & 0 & 0.428288 & 0 & 0 & 0 & 0 \\
0 & 0 & -1.25078 & 0 & 0 & -0.468106 & 0 & -0.0984261 & 0.0784189 & 0 & 0 \\
0 & 0 & -0.427975 & 0 & 0 & 0.650154 & 0 & -0.500656 & 0.00594322 & 0 & 0 \\
0 & 0 & 0.0933032 & 0 & 0 & 0.111285 & 0 & 0.0796181 & 0.606001 & 0 & 0 \\
-3.44464 & -1.02341 & 0 & 0 & 0 & 0 & 0.724775 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0.106061 & -0.0597173 & 0 & 0 & 0 & 0 & 0.529804 & 0.202300 \\
0 & 0 & 0 & -0.609665 & -1.08279 & 0 & 0 & 0 & 0 & -0.0198155 & 0.0518949 \\
0 & 0 & 0 & 0.0597173 & 0.106061 & 0 & 0 & 0 & 0 & -0.202300 & 0.529804 \\
0 & 0 & 0 & 1.08279 & -0.609665 & 0 & 0 & 0 & 0 & -0.0518949 & -0.0198155
\end{pmatrix}$$

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